

Anomaly Detection in Industrial Components

Exploratory Data Analysis Report

We analysed the MVTec AD dataset for our project “Anomaly Detection in Industrial Components”. The report focuses on understanding the data and the properties required for future preprocessing and evaluation, aligned with the benchmarking standards introduced in “AUPIMO: Redefining Visual Anomaly Detection”. The main report presents a clear and condensed narrative, while detailed calculations, integrity checks, and additional figures are provided in the appendix.

1. Dataset composition and annotations

Our dataset contains 15 categories of 2D images of different items. Each category's images are already pre-split into a training set containing only non-anomalous (good) images, a test set containing anomalous and good images, and a ground-truth set containing masks that mark the locations of anomalies.

Sample of defects with mask per product

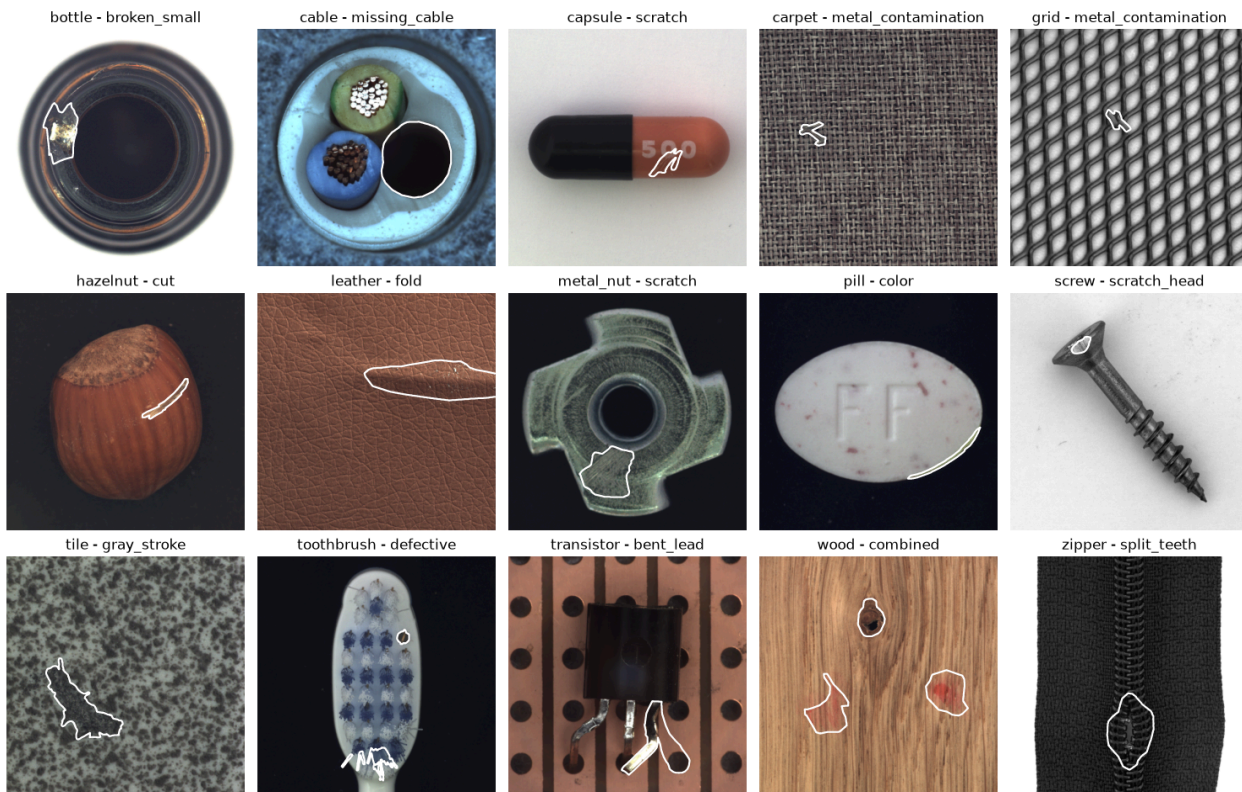


Figure 1: Sample set of anomalous images with applied ground-truth masks

Fig. 1 displays anomalous samples overlaid with ground truth masks to mark the location of anomalies. The placement of the sets under any anomaly category assigns them the specific label for binary anomaly detection, while the masks supply pixel-level spatial annotations for anomaly localization. The images help to understand the precision and structure of the masks and give a visual impression of the size of different anomalies relative to the size of depicted industrial components. The structural complexity and detailed nature of the ground truth annotations demonstrate that the modeling architecture must maintain high-resolution spatial awareness. Reduced image resolution during feature extraction may risk blurring small anomaly boundaries.

2. Pixel-level imbalance and defect-size distribution

An inherent property of the dataset is the imbalance between good and anomalous images as well as the imbalance between defective and good pixels. The training set contains only good images, and even within defective test images most pixels are normal. The median defect occupies approximately 1.54% of the image, while 25% of defects occupy less than approximately 0.49%. At the pixel-level, this class imbalance leads to an accuracy paradox. Consequently, for model training and evaluation, it is paramount to use the area under the precision-recall curve and the corresponding F_β -values to assess the model performance.

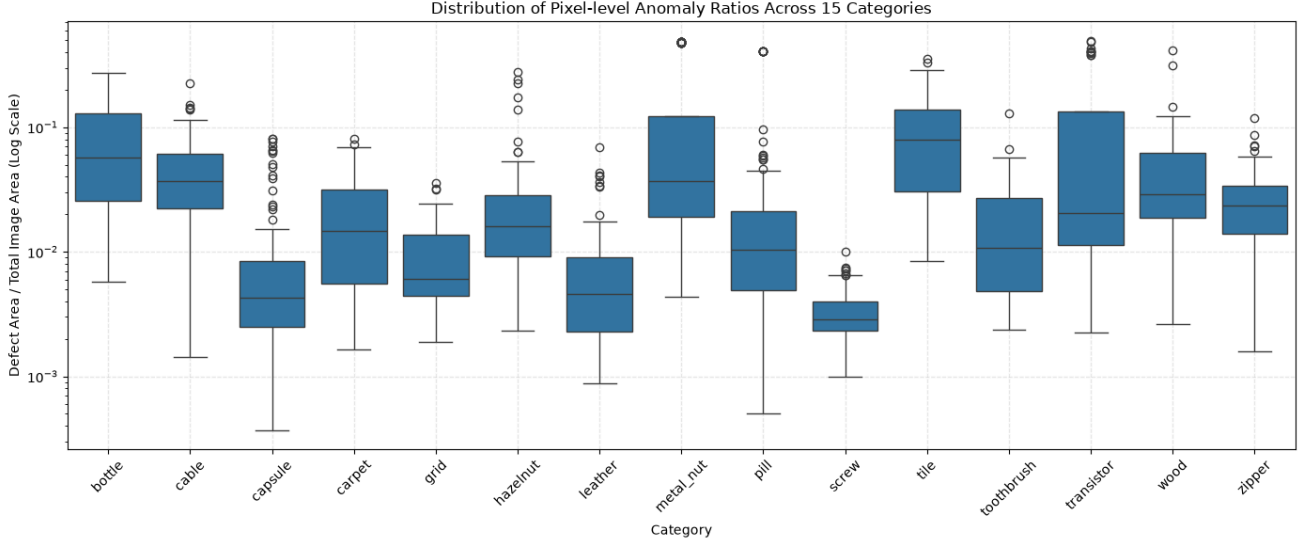


Figure 2: Distribution of pixel-level anomaly ratios across the 15 categories

Based on the results, we can see that there is an imbalance in the test set between defect-free and defective samples, and the degree of imbalance varies by product type. We established that the dataset is highly imbalanced at the pixel-level, different physical categories present completely different defect size distributions.

For the statistical proof, we perform the Kruskal-Wallis H-Test: We test the null hypothesis that the population medians of the image-level defect ratios are equal across all 15 categories. We choose Kruskal-Wallis test over a pixel-level Chi-Square test because:

- Pixels are highly spatially correlated (violating the independence assumption required for Chi-Square).
- The massive pixel sample size ($N > 67$ million) artificially inflates the Chi-Square power, causing even trivial, microscopic differences to appear statistically significant.
- The Kruskal-Wallis test treats each anomalous image ($N = 1258$) as a single independent observation, making the statistical test trustworthy.

The test gives as result for $H = 626.5$, where $H = 0$ would mean that categories are basically identical in the defect size distribution and our result points to a significant imbalance. Further, our resulting p-value $p = 1.2e-124$ tells us with almost certainty to reject the null hypothesis and accept instead that the distribution of defect pixel ratios significantly varies across categories. The result tells us that it is unlikely to find a single binarization threshold that performs well across all categories.

To understand the intra-category difficulty, Fig. 3 breaks down defect size distributions by specific defect modes. The separation in scale across defect types within the same product categories demonstrates that model failure modes will likely be mode-specific; thus, evaluation performance may vary by defect type or prevent large, obvious defects from masking poor performance on subtle anomalies.

Defect-size variation by defect type

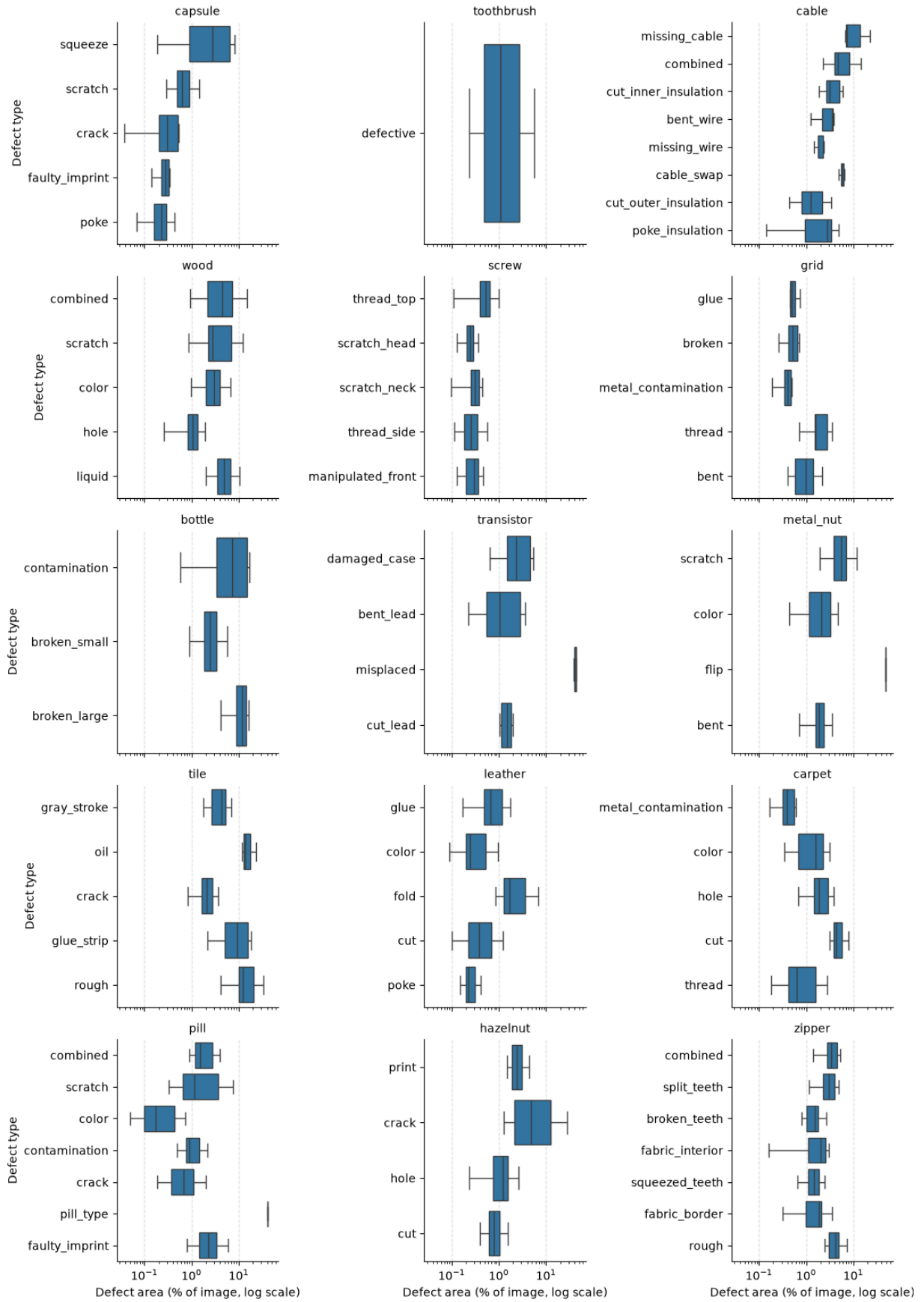


Figure 3: Defect-size distributions reveal intra-category and defect-mode variance

It can be concluded that defect sizes vary not only between products but also between defect modes of the same product. Some defect types are consistently small, while others cover a much larger part of the object.

Later evaluation should report results per product and, where sample size allows, per defect type or defect-size group.

3. Defect shape and complexity

We combine the analysis of defect size with defect shape and complexity. Ground-truth masks differ in bounding-box fill, number of connected components and contact with image borders.

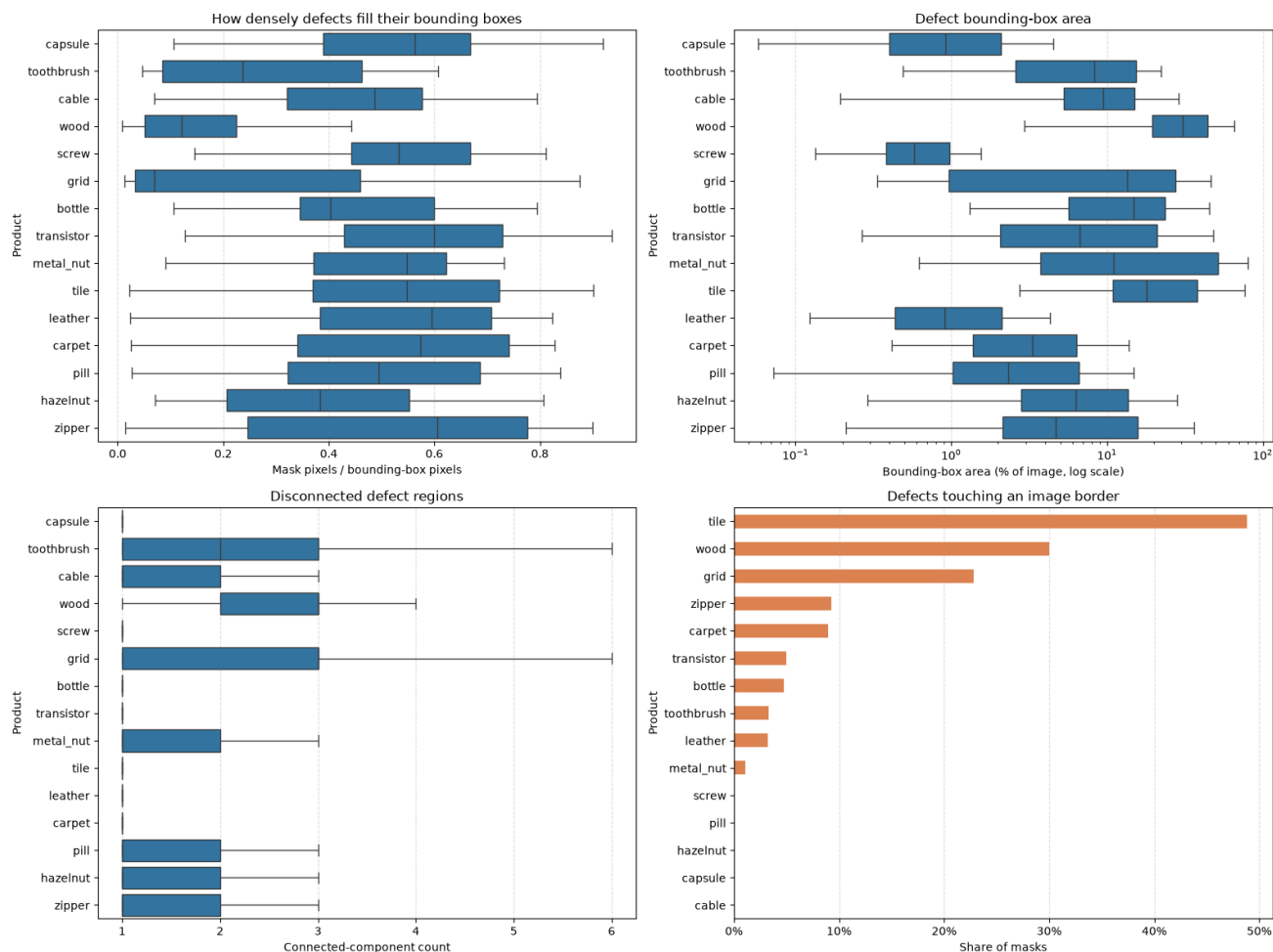


Figure 4: Defect shape complexity, fragmentation and border contact across categories

Defects vary widely in shape and fragmentation. Some are thin and occupy only a small part of their bounding box, some consist of several disconnected regions, and several categories contain defects touching the image border.

Later preprocessing should avoid transformations that remove border regions or erase thin structures. Pixel-level localization is more informative than using defect size or a bounding box alone.

4. Defect spatial locations

Mapping defect centre-of-mass coordinates to a normalized [0, 1] coordinate system shows that some categories have broad spatial dispersion while others exhibit stronger positional clustering. This is a useful context, but location should not be treated as proof of defectiveness.

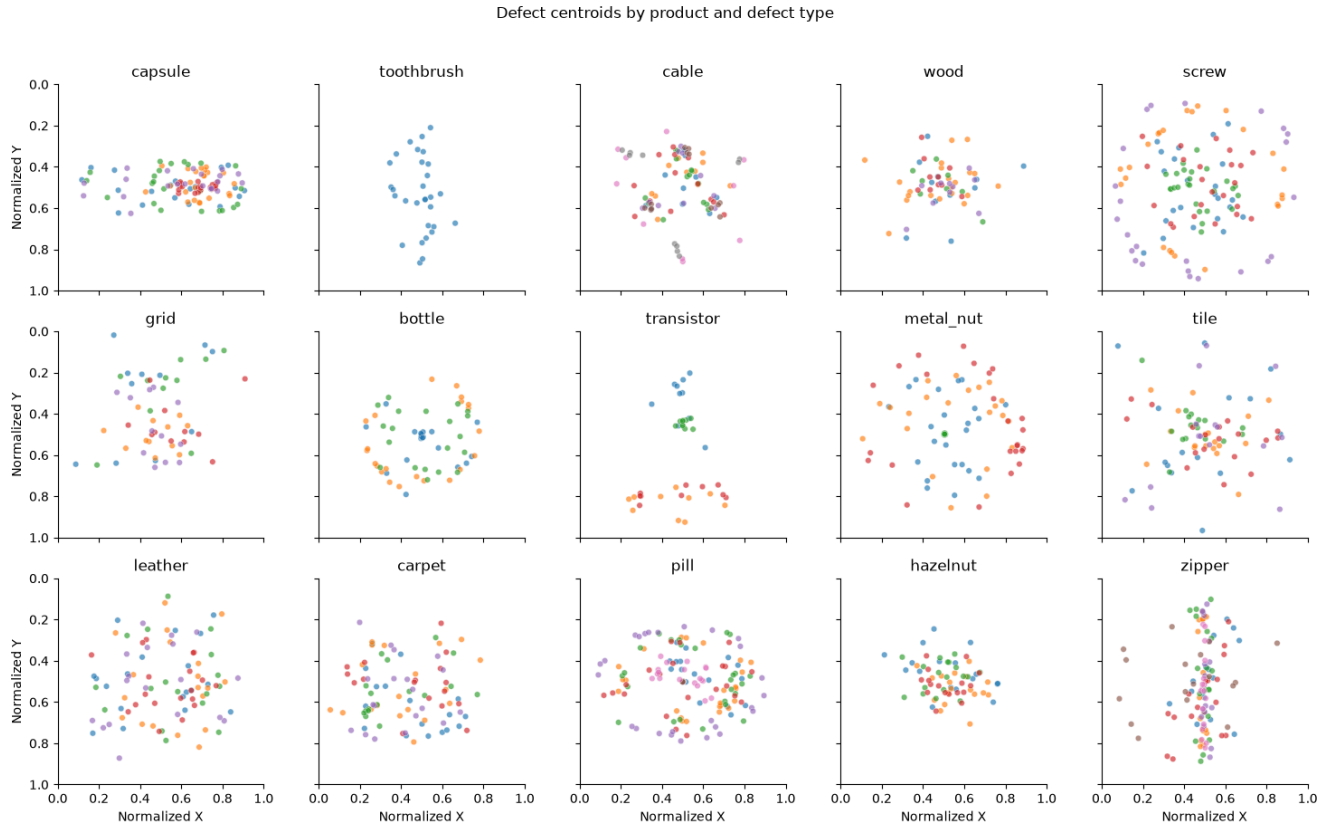


Figure 5: Spatial mapping of defect centroids by category using normalized axes

Spatial distributions of defects differ by product. Some rigid-object categories show concentrated defect locations, while several texture categories show broader dispersion. These patterns describe positional bias in the dataset, not physical rules that defects can never occur elsewhere. If a model predicts a defect in the top-left corner of a bottle image, it is likely to be false positive, because the data proves defects simply do not physically occur there.

Spatial information may help with aligned rigid objects, but models must be checked for positional shortcuts. Augmentation and evaluation should ensure that performance reflects visual anomaly detection rather than memorizing common defect locations.

5. Normal train/test brightness and contrast

To determine whether the predefined train/test split contains acquisition differences, we compare brightness and contrast within each category using normal training and normal test images. These global statistics are indicators of possible covariate shift; they are not direct detectors of localized defects.

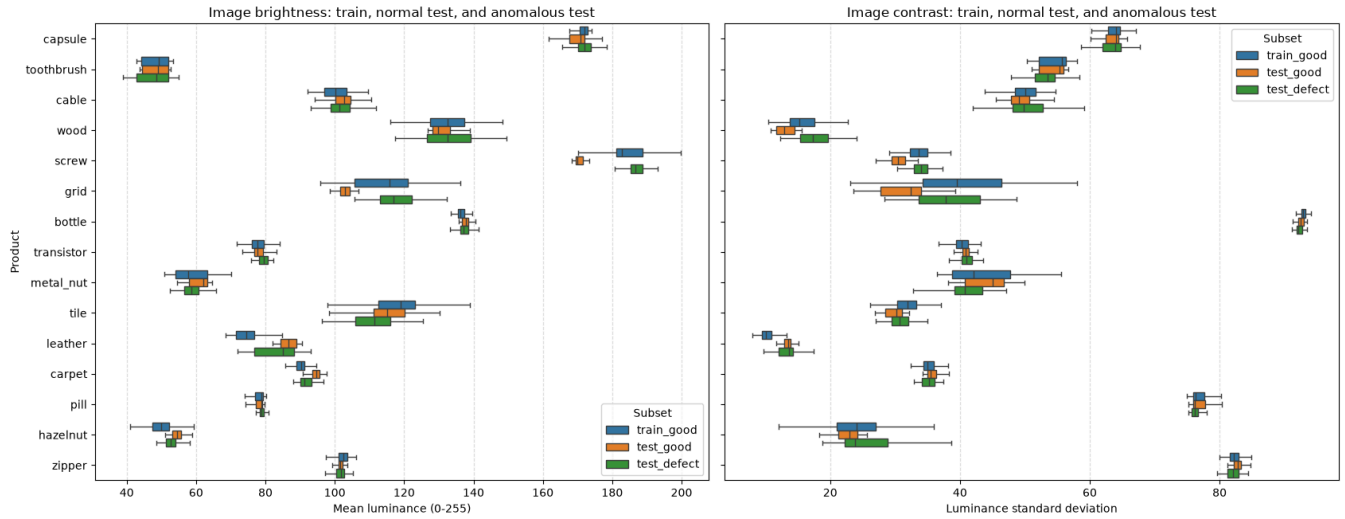


Figure 6: Brightness and contrast distributions across normal train and test images by category

Some categories, particularly screw, leather and grid, show noticeable differences in brightness or contrast between normal train and test images, while other categories are comparatively stable.

A model may otherwise interpret acquisition differences as anomalies. Preprocessing and model validation should therefore be considered per product, while avoiding normalization that removes genuine defect signals.

6. Conclusions and implications

- **Dataset structure:** Training images are good-only, while test composition and defect-type counts vary by product category. Results should therefore be reported per product rather than only as a global average.
- **Defect scale, morphology and location:** Most defects are small, and masks vary in shape, fragmentation, border contact and spatial location. Some categories show positional clustering, while others are broadly distributed. These findings support spatially sensitive analysis while warning against models that rely on location shortcuts.
- **Normal-domain shift:** Brightness and contrast differences between normal train and test images may generate false alarms if a model learns acquisition conditions instead of normal product structure.
- **Limited training volume:** Each category contains only approximately 200 to 300 normal training images. Training a deep neural network from scratch may overfit; transfer learning with pretrained features should be considered in the modeling phase.
- **Evaluation implication:** Accuracy is misleading under severe imbalance. Pixel-level AUROC has a related weakness because the very large number of normal pixels can keep the false-positive rate small even when many normal pixels are incorrectly flagged. Average Precision or PR-AUC should be the primary imbalance-sensitive metric; AUROC may still be reported for comparison with published work.

Appendix

This appendix preserves supporting checks, calculations and supplementary evidence without interrupting the main narrative.

A. Dataset integrity and format checks

- Dataset manifest and mask integrity: A manifest was built listing every image and mask, including product, split, defect type, dimensions and mask path. Every defective test image has a mask, image-mask pairing is complete, paired dimensions match and all masks are grayscale.
- Exact duplicates and train/test leakage: SHA-256 hashes were used to identify byte-for-byte duplicates. No exact duplicate images or exact train/test leakage were found.
- Image modes and dimensions: Most images are RGB, while grid, screw, tile, toothbrush and zipper are grayscale. Image dimensions vary by product, with widths and heights between 700 and 1024 pixels; the inspected images remain square.
- Dataset volume: The dataset contains 3,629 normal training images, 467 normal test images and 1,258 defective test images with corresponding masks.

B. Detailed defect-area statistics

Statistic	Defect area (% of image)	Interpretation
Count	1,258	One observation per defective image
Mean	4.39%	Raised by a small number of very large defects
Standard deviation	8.74%	Strong geometric variability
Minimum	0.037%	Extremely small defect
25th percentile	0.49%	One quarter occupy less than 0.5%
Median	1.54%	Typical defect occupies a small image fraction
75th percentile	3.89%	Three quarters occupy less than 4%
Maximum	49.32%	Large structural defect covering nearly half the image

C. Statistical tests

C.1 The Kruskal-Wallis H-test:

It compares the distributions of image-level defect ratios across the 15 categories. It was preferred over a pixel-level chi-square test because neighbouring pixels are spatially correlated, the very large number of pixels would make even negligible differences statistically significant, and the image is the more defensible observational unit.

- Null hypothesis: the category distributions of image-level defect ratios are identical.
- H-statistic: 626.507
- p-value: 1.21×10^{-124}
- Conclusion: reject the null hypothesis. Defect-size distributions differ significantly across categories. This result does not by itself prove that a universal threshold must fail, but it supports category-stratified evaluation.

C.2 Pixel-level Chi-Square test:

Null hypothesis: all categories share the same underlying anomalous-pixel ratio.

Chi-square statistic: $\chi^2(14) \approx 41.7$ million, Degrees of freedom: 14, p-value: $p \approx 0$

Conclusion: reject the null hypothesis. Anomalous-pixel ratios range from 0.34% (screw) to 14.5% (metal_nut), a 43-fold spread. As with the Kruskal-Wallis result above, at this sample size ($N > 67$ million) the effect size is more meaningful than the p-value alone.

Note: this test proves a different claim than Kruskal-Wallis, not the same one twice. Kruskal-Wallis shows that defect size varies across categories (per-image ratios have different medians). The Chi-Square test shows that the overall proportion of anomalous pixels varies across categories (category-level totals differ).

D. Why AUROC should not be used for pixel-level evaluation

Two questions guide this decision, following the same reasoning we used for accuracy in Section 2.

First: what do published benchmarks tell us?

Published MVTec AD results show AUROC is close to saturation: PatchCore and SimpleNet report 99.6%, EfficientAD up to 99.8%, Dinomaly 99.6%. When top methods differ by only a few tenths of a percent, AUROC cannot reliably distinguish model quality.

Second: does AUROC have the same weakness as accuracy, using our own data? Yes.

Our own pixel-level analysis (Section 2) showed only ~1.5% of pixels in a defective image are anomalous, meaning ~98.5% are normal. Accuracy fails here because predicting everything normal still yields ~98.5% accuracy while catching zero real defects.

The exact same flaw applies to AUROC, which is built on $FPR = FP / (FP + TN)$.

Since true negatives dominate the denominator (~98.5% of pixels), a model can misclassify many pixels while FPR barely moves. This keeps AUROC deceptively high (often above 95%) even with poor real performance. AUROC also averages across every possible threshold, including ones that would never be used in a real low-false-alarm factory setting.

In short: AUROC gets washed out by the huge pool of easy, ignored normal pixels, the same way accuracy gets artificially raised by them. Precision does not have this problem, since it only looks at pixels the model actually flags:

$$\text{Precision} = TP / (TP + FP)$$

Precision can't hide behind the large pool of correctly-ignored normal pixels, since those pixels never enter its calculation, making it a more honest score under this level of imbalance.

Conclusion: Because AUROC is uninformative under severe pixel imbalance, it is discarded for evaluation. We use pixel-level Average Precision (PR-AUC) and AUPIMO as our evaluation benchmarks. Results will be reported separately per product and per level (image vs. pixel).

E. Dataset Composition

One-class training and uneven test imbalance

